

2 The Broken Toy

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Session 1

One-line motivation. *Raw coordinates are a projection. They mix causes, hide structure, and mislead algorithms. Before we can learn anything useful, we must ask what representation the phenomenon actually deserves.*

Session 0 gave us the projectile. The failure there was that our *hypothesis class* could not express the shape of the phenomenon. The cantilever is a different failure with the same root cause: the *coordinates themselves* mix the causes. In both cases, the fix was to change the representation, and in both cases, no amount of data or tuning could have saved the original setup.

2.1 The Phenomenon

We begin with a question that has nothing to do with machine learning, and everything to do with it.

What are these two tables in Figure 3?

To a computer, they are nothing but two grids of real numbers. Each entry is a pixel intensity, a raw sensor reading, a single coordinate in a high-dimensional space.

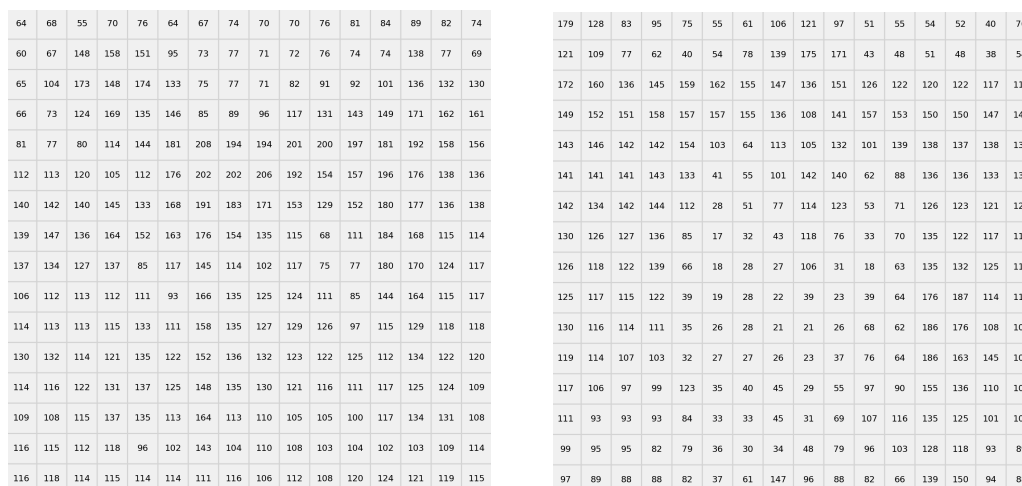


Figure 3: Two raw images as arrays of pixel intensities. On the left, a dog. On the right, an anthropoid. As numbers, they are structurally indistinguishable.

At first glance the two arrays look indistinguishable. They are the same size, they have the same range of values, and if you shuffled their entries you could not tell which one came from which. Yet when we look at them, we do not see numbers. We see a dog and an anthropoid. Nothing in the raw numbers themselves announces “dog” or “anthropoid”. The meaning lives entirely in the *pattern*—in how the numbers are arranged relative to one another.

What machine learning does, at its core, is learn to read such patterns. It takes a grid of numbers and produces a label, a category, a decision. And remarkably, it can do this even when the underlying structure is invisible to us in the raw form. The gap between the numbers and the meaning is exactly the gap that learning algorithms are designed to bridge.



Figure 4: The same two arrays, now rendered in color. Structure that was invisible in the raw numeric form—edges, contours, regions—becomes visible. The numbers have not changed. Only the representation has.

When we render the same arrays in color—mapping each numeric value to a hue—the underlying structure becomes visible. Regions that share similar values light up with similar colors; edges and contours emerge from the noise. The numbers have not changed. Only our representation of them has. This is a small but important lesson: *the same data can look meaningless or*

meaningful depending entirely on how we choose to present it, and much of the art of machine learning lies in choosing representations that make the hidden structure visible.

*The raw representation is semantically empty. The same numbers could be a Trump!, a man, a galaxy, or noise. **Richness of representation is what makes meaning possible.***

A dog and Trump are distinguishable in the world. They are not distinguishable in their raw pixel coordinates. The gap between the phenomenon and the data is where machine learning lives.

But recognizing this gap is not enough. The real question—the one that will occupy us for the rest of the course—is this:

***How can we manipulate data with respect to its structure?** How do we transform a raw grid of numbers into a representation in which the structure we care about is not merely present, but visible, usable, and connected to a decision?*

2.2 The Broken Toy: A Cantilever Beam

Now we make the question concrete. Consider a cantilever beam with two strain gauges at the same cross-section:

- one on the **top surface**, reading ε_{top} ,
- one on the **bottom surface**, reading $\varepsilon_{\text{bottom}}$.

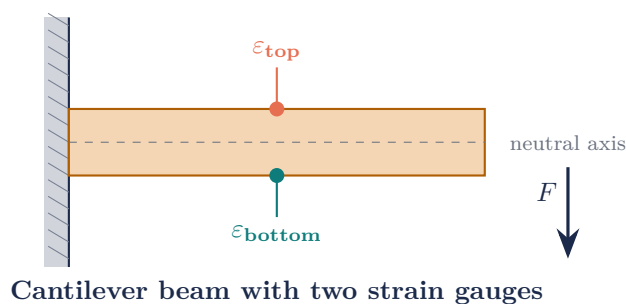


Figure 5: A cantilever beam with two strain gauges. The neutral axis experiences no bending strain.

The beam can experience two independent physical effects:

- **Axial strain** ε_a : tension or compression. Same sign on top and bottom.
- **Bending strain** ε_b : bending. Opposite signs on top and bottom.

The raw sensor vector is

$$\mathbf{x}_{\text{raw}} = \begin{bmatrix} \varepsilon_{\text{top}} \\ \varepsilon_{\text{bottom}} \end{bmatrix}.$$

But the physically meaningful coordinates are

$$\varepsilon_a = \frac{\varepsilon_{\text{top}} + \varepsilon_{\text{bottom}}}{2}, \quad \varepsilon_b = \frac{\varepsilon_{\text{top}} - \varepsilon_{\text{bottom}}}{2}.$$

These coordinates come from the mechanics of beam deformation, not from the data. They are supplied by physics, in the same way that the sinusoidal basis in Session 0 was supplied by the physics of projectile motion. Finding the right representation is not a purely data-driven task; it requires prior knowledge about the phenomenon.

In matrix form:

$$\begin{bmatrix} \varepsilon_{\text{top}} \\ \varepsilon_{\text{bottom}} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} \varepsilon_a \\ \varepsilon_b \end{bmatrix} + \boldsymbol{\eta}, \quad (2)$$

where $\boldsymbol{\eta}$ is measurement noise.

Remark 2.1: What Equation (2) Says

The raw measurement is a *rotated and scaled* view of the true physical state. The matrix $\mathbf{A}$ is invertible, so no information is lost in principle. But the coordinates are *mixed*: each raw reading depends on both physical causes.

2.3 The Failure

Suppose our task is to detect bending. We collect N samples, label each by whether $\varepsilon_b > 0$, and train a shallow decision tree on the raw coordinates $(\varepsilon_{\text{top}}, \varepsilon_{\text{bottom}})$.

The tree partitions the plane with axis-aligned splits. But the physical boundary $\varepsilon_b = 0$ corresponds to

$$\varepsilon_{\text{top}} - \varepsilon_{\text{bottom}} = 0 \quad \Longleftrightarrow \quad \varepsilon_{\text{top}} = \varepsilon_{\text{bottom}},$$

which is a *diagonal* line in the raw plane.

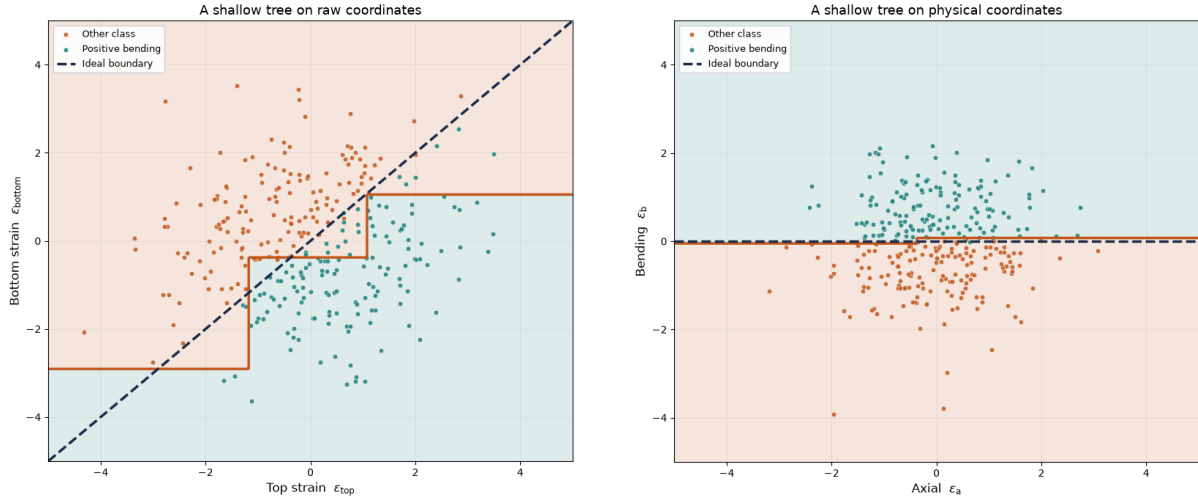
A decision tree cannot represent a diagonal boundary without a staircase of many axis-aligned splits. The model is not stupid. **The coordinates are stupid.**

2.4 The Representational Shift

Now rotate the coordinates:

$$\text{axial} = \frac{\varepsilon_{\text{top}} + \varepsilon_{\text{bottom}}}{2}, \quad \text{bending} = \frac{\varepsilon_{\text{top}} - \varepsilon_{\text{bottom}}}{2}.$$

In these coordinates, the bending problem becomes a one-dimensional threshold. The same model family, with the same capacity, now works with almost no effort.



(a) Raw coordinates: axis-aligned splits struggle to isolate the physical direction.

(b) Physical coordinates: the same model family separates the classes trivially.

Figure 6: Left: raw coordinates force axis-aligned approximations. Right: the same data in physical coordinates is trivially separable.

The model did not get smarter. The representation got honest. Raw coordinates are not the problem—they are a coordinate system, and it was the wrong one.

2.5 Geometry and Intuition

What just happened, geometrically?

- The physical state lives in a 2D space spanned by $(\varepsilon_a, \varepsilon_b)$.
- The observation map is a linear transformation $\mathbf{A}$ plus noise.
- The task depends only on ε_b , i.e., on one coordinate in the physical basis.
- In the raw basis, this coordinate is a linear combination of both measurements.

Definition 2.1: Change of Basis

Let $\mathbf{x} \in \mathbb{R}^n$ be a raw observation and let $\mathbf{W} \in \mathbb{R}^{n \times n}$ be invertible. The representation $\mathbf{z} = \mathbf{W}\mathbf{x}$ is a change of basis. The structure of the problem is preserved, but its *visibility* to a given model class may change dramatically.

2.6 From the Physical World to Raw Data

Before any learning can happen, we must collect data from the physical world. Let $\mathcal{W}$ denote the continuous physical reality. Computers cannot process infinite-dimensional systems directly, so sensors, cameras, or measurement devices discretize $\mathcal{W}$ into a finite dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \subset \mathcal{X} \times \mathcal{Y}$$

where $x_i \in \mathbb{R}^D$ is a feature vector of raw measurements and $y_i \in \mathcal{Y}$ is the corresponding target. Observation is never exact. Sensor precision, unobserved variables, and environmental noise all corrupt the measurement. We never see the true relationship $f(x)$ directly; we only see noisy samples.

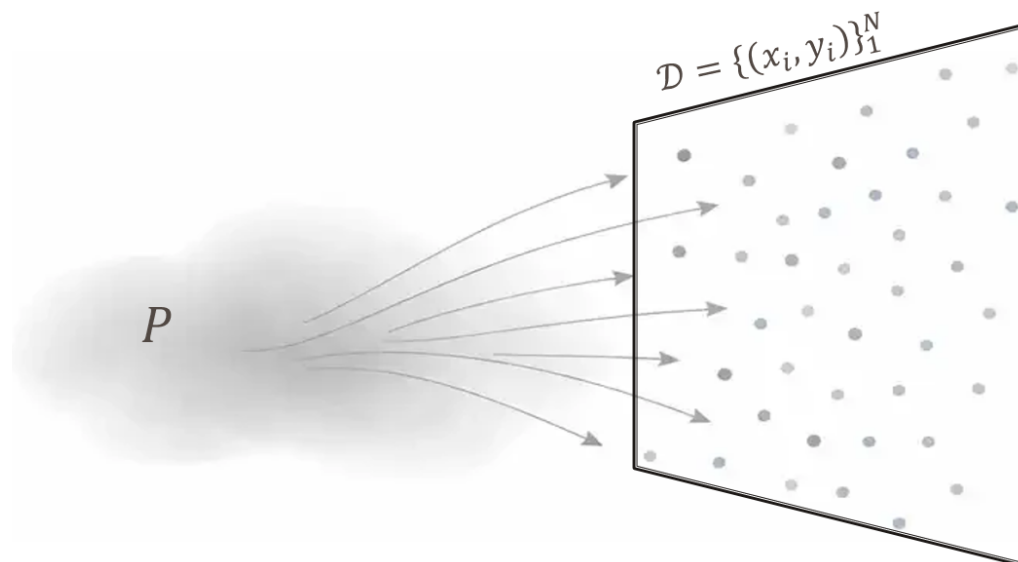


Figure 7: Continuous physical world $\mathcal{W}$ projected onto a discrete observation plane $\mathcal{D}$.

A digital photograph is the simplest example: the continuous scene is discretized into a grid of pixels, each with finite intensity. Learning happens on these discrete points, not on continuous reality.

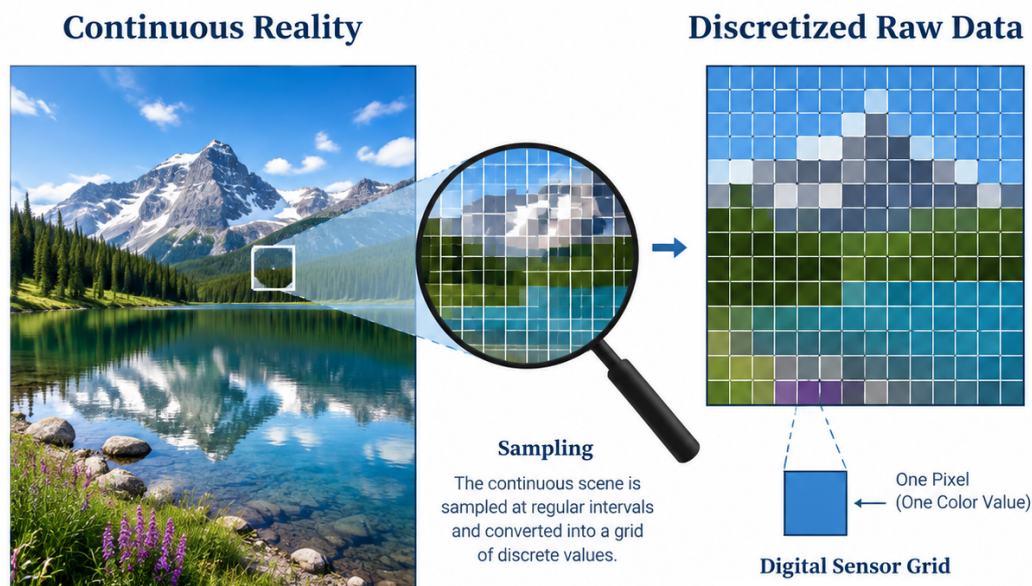


Figure 8: Continuous reality to discretized data.

2.7 The Representation Pipeline

Every problem in this course follows the same ontology. A learning system is a composition of two maps:

1. **Physical Observation Mapping $\mathcal{O}$** : the environment P is sampled to form the raw dataset $\mathcal{D}$.

$$\mathcal{O} : P \longrightarrow \mathcal{D} \quad (3)$$

2. **Representation Mapping $\mathcal{A}$** : a learning algorithm maps raw features into a learned secondary representation space $\mathcal{Z}$.

$$\mathcal{A} : \mathcal{D} \longrightarrow \mathcal{Z} \quad (4)$$

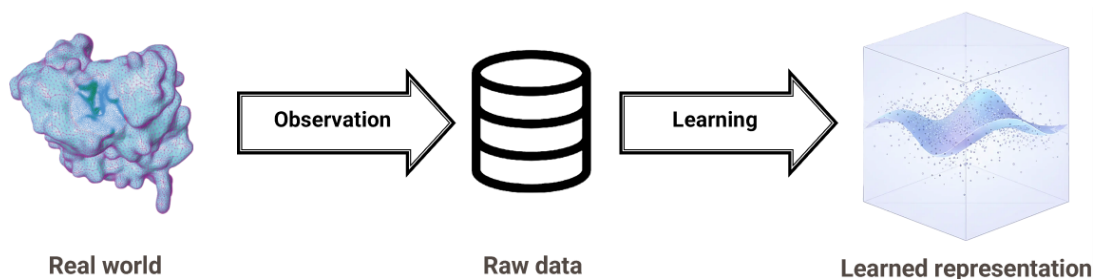


Figure 9: A learning system is a composition of two maps ($\mathcal{W} \xrightarrow{\mathcal{O}} \mathcal{D} \xrightarrow{\mathcal{A}} \mathcal{Z}$).

The phenomenon has structure, symmetry, and useful information *before* we ever observe it. Observation is a lossy projection. The algorithm is a second projection. The secondary representation is meaningful only if it preserves what the first projection did not destroy.

Definition 2.2: The Representation Problem

Given a phenomenon P with structure $\mathcal{S}$, an observation map $o : P \rightarrow \mathbb{R}^n$, and a task T , find a representation $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that the structure relevant to T is preserved or made explicit in the coordinates $\phi(o(P))$.
The algorithm $\mathcal{A}$ is the procedure that finds ϕ .

The rest of this course is the systematic study of $\mathcal{A}$.

2.8 Translation to Code

The code that generates the cantilever data and reproduces every figure in this section is in the live notebook:

Session 01 — Live Notebook

In the notebook you will find:

- the data generator (axial, bending, noise);
- the scatter plots in raw and physical coordinates;
- the shallow decision tree comparison and its cross-validation scores.

The two representations are built by the same change of basis:

$$\varepsilon_a = \frac{\varepsilon_{\text{top}} + \varepsilon_{\text{bottom}}}{2}, \quad \varepsilon_b = \frac{\varepsilon_{\text{top}} - \varepsilon_{\text{bottom}}}{2}.$$

2.9 What the Cantilever Taught Us

Let us be careful about what we have actually shown: *when the coordinates mix independent causes, and the model has a bias toward axis-aligned structure, the failure is representational, not algorithmic.*

This gives us a specific diagnostic for the cantilever’s class of problem:

1. What structure does the phenomenon have?
2. What does the observation preserve, and what does it mix?
3. Does the model’s assumptions align with that structure?

If the answer to (3) is no, the model will fail. This is the shape of the failure the cantilever has shown us. In Session 2 we will generalize this into a diagnostic habit that applies to a much wider range of problems.

2.10 The Same Lesson, Twice

Sessions 0 and 1 have now shown the same root cause through two different mechanisms. The projectile’s hypothesis class could not express the shape of the phenomenon. The cantilever’s coordinates mixed the causes. Both are representation failures. Both were fixed by changing the representation. Both would have been immune to any amount of data or tuning.

	Projectile (Session 0)	Cantilever (Session 1)
Wrong representation	raw (θ, R)	raw $(\varepsilon_{\text{top}}, \varepsilon_{\text{bottom}})$
Failure mode	hypothesis class too simple	coordinates mixed independent causes
Fix	change the features	change the basis
Where the fix came from	Newtonian mechanics	mechanics of beam deformation

Table 1: Two problems, two failures, one cause.

2.11 Connection to Next Session

We have seen that raw coordinates can be misleading, and that a change of basis can rescue a model. But notice something uncomfortable about both examples: in Session 0, physics supplied the shape. In Session 1, physics supplied the coordinates. Both times, we knew the answer before we looked at the data.

What happens when we do not? What happens when we have measurements from a system whose physics we do not understand, and we cannot write down the correct coordinates in advance? Do we have any hope of learning the representation from the data itself?

This is the question of Session 2: **The Hidden Shape.**

2.12 Exercises

1. **The inverse observation map.** The measurement model for the cantilever is

$$\begin{pmatrix} \varepsilon_{\text{top}} \\ \varepsilon_{\text{bottom}} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} \varepsilon_a \\ \varepsilon_b \end{pmatrix} + \begin{pmatrix} \eta_{\text{top}} \\ \eta_{\text{bottom}} \end{pmatrix}.$$

Derive $\mathbf{A}^{-1}$ explicitly. Show that, in the noiseless case ($\eta = 0$), the axial and bending components are recovered exactly from the raw readings as

$$\varepsilon_a = \frac{\varepsilon_{\text{top}} + \varepsilon_{\text{bottom}}}{2}, \quad \varepsilon_b = \frac{\varepsilon_{\text{top}} - \varepsilon_{\text{bottom}}}{2}.$$

Why does the factor of $1/2$ appear in both formulas? What geometric operation does each formula perform on the raw coordinate plane?

2. **Removing measurement noise.** Set $\eta = 0$ in the model above. Regenerate the cantilever data with no noise and retrain the depth-3 decision tree on the raw coordinates $(\varepsilon_{\text{top}}, \varepsilon_{\text{bottom}})$. Compare its accuracy to the noiseless physical coordinates $(\varepsilon_a, \varepsilon_b)$. Does the accuracy gap between the two representations close, stay the same, or grow? Explain the result in terms of the decision boundary, not the data.
3. **Changing the model family.** The shallow decision tree and logistic regression encode different geometric biases. Replace the tree with a logistic regression classifier and repeat the comparison on both representations. Report the accuracy for each. Does the gap between raw and physical coordinates shrink, remain, or grow? Draw the decision boundary that logistic regression learns in each coordinate system, and explain the result using the geometry of that boundary.